

2006 级概率论与数理统计试卷 A 卷参考答案

一、

1.D

注释: $P(X \leq 1+u) = P\left(\frac{X-u}{\sigma} \leq \frac{1+u-u}{\sigma}\right) = \Phi\left(\frac{1}{\sigma}\right)$

2.C

注释: 参考课本第 8 页

3.A

注释: 连续型随机变量在某一个点上的概率取值为零, 故 A 正确

? B 项是否正确

4.B

注释: 参考课本 86 页

5.A

二、

1. 1.33(或者填 $\frac{1359}{1024}$)

2. 25

注释: 参考课本 86 页

3. 0.25

4. $(X+Y) \sim B(7, p)$

注释: $E(X)=3p, E(Y)=4p$, 故 $E(X+Y)=E(X)+E(Y)=3p+4p=7p$;

$D(X)=3p(1-p), D(Y)=4p(1-p)$ 且 X, Y 独立, 故 $D(X+Y)=D(X)+D(Y)=3p(1-p)+4p(1-p)$

设 $(X+Y) \sim B(n, P)$, 则有
$$\begin{cases} E(X+Y)=7p=nP \\ D(X+Y)=3p(1-p)+4p(1-p)=nP(1-P) \end{cases}$$

解得 $n=7, P=p$

5. 2/5

X 的密度函数为 $f(x) = \frac{1}{5}$

方程有实根，则必须满足 $\Delta = b^2 - 4ac = (2X)^2 - 4 \times 1 \times (5X - 4) \geq 0$ ，
即 $X \leq 1$ 或者 $X \geq 4$ 。

故方程有实根的概率 $P = \int_0^1 \frac{1}{5} dx + \int_4^5 \frac{1}{5} dx = \frac{2}{5}$

6. 0.35

由 $E(3X + 5) = 11$ 得 $EX = 2$

由 $D(3X + 5) = \sigma^2$ 得 $DX = \frac{\sigma^2}{9}$, $\sqrt{DX} = \frac{\sigma}{3}$

因 $P\{2 < X < 4\} = 0.15$, 故 $P\{\frac{2-2}{\frac{\sigma}{3}} < \frac{X-2}{\frac{\sigma}{3}} < \frac{4-2}{\frac{\sigma}{3}}\} = 0.15$

所以 $\Phi\left(\frac{2}{\frac{\sigma}{3}}\right) - \Phi(0) = 0.15$

所以 $\Phi\left(\frac{2}{\frac{\sigma}{3}}\right) - \Phi\left(\frac{-2}{\frac{\sigma}{3}}\right) = 0.3$

$P\{X < 0\} = P\left\{\frac{X-2}{\frac{\sigma}{3}} < \frac{0-2}{\frac{\sigma}{3}}\right\} = \Phi\left(\frac{-2}{\frac{\sigma}{3}}\right) = [1 - \Phi\left(\frac{2}{\frac{\sigma}{3}}\right) - \Phi\left(\frac{-2}{\frac{\sigma}{3}}\right)]/2$
_____ = $[1 - 0.3]/2 = 0.35$

7. 相关

三、

设 A = “取出的产品是正品”；

$B_{甲}$ = “取出的产品是甲厂生产的”

$B_{乙}$ = “取出的产品是乙厂生产的”

$B_{丙}$ = “取出的产品是丙厂生产的”

则 $P(A) = P(AB_{甲}) + P(AB_{乙}) + P(AB_{丙})$

$= 0.5 \times 0.9 + 0.3 \times 0.8 + 0.2 \times 0.7 = 0.83$

$P(B_{甲} | A) = \frac{P(AB_{甲})}{P(A)} = \frac{P(B_{甲}) \cdot P(A | B_{甲})}{P(A)} = \frac{0.5 \times 0.9}{0.83} = 0.54$

四、

$$X \sim \begin{pmatrix} -1 & 1 & 3 \\ 0.3 & 0.5 & 0.2 \end{pmatrix}$$

$$EX = (-1) \times 0.3 + 1 \times 0.5 + 3 \times 0.2 = 0.8$$

五、

$$(1) \text{由题意} f(x) = \begin{cases} \frac{1}{2}e^x & x \leq 0 \\ \frac{1}{2}e^{-x} & x > 0 \end{cases}$$

$$\text{故} F(x) = \begin{cases} \frac{1}{2}e^x & x \leq 0 \\ 1 - \frac{1}{2}e^{-x} & x > 0 \end{cases}$$

$$(2) P(-5 < X < 10) = (1 - \frac{1}{2}e^{-10}) - (\frac{1}{2}e^{-5})$$

$$(3) EX = \int_{-\infty}^0 x \cdot \frac{1}{2}e^x dx + \int_0^{+\infty} x \cdot \frac{1}{2}e^{-x} dx = \frac{1}{2}[(x-1)e^x]_{-\infty}^0 + \frac{1}{2}[(-x-1)e^{-x}]_0^{+\infty} \\ = 0$$

$$EX^2 = \int_{-\infty}^0 x^2 \cdot \frac{1}{2}e^x dx + \int_0^{+\infty} x^2 \cdot \frac{1}{2}e^{-x} dx$$

$$= \frac{1}{2}[x^2 e^x - 2x e^x + 2e^x]_{-\infty}^0 + \frac{1}{2}[-x^2 e^{-x} - 2x e^{-x} - 2e^{-x}]_0^{+\infty} = 2$$

$$DX = EX^2 - (EX)^2 = 2$$

? 六、

设 X_i 为第 i 台机床生产的次品率

$$X_i \sim U(0.005, 0.035)$$

$$EX_i = \frac{0.005 + 0.035}{2} = 0.02, DX_i = \frac{1}{12}(0.035 - 0.005)^2 = 0.000075$$

(注: 对于均匀分布 $X_i \sim U(a, b)$, 有 $EX_i = \frac{a+b}{2}$, $DX_i = \frac{1}{12}(b-a)^2$)

$$\text{设总次品率 } Y = \sum_{i=1}^{2000} X_i$$

若要满足这批产品的平均次品率小于0.025, 则 $Y < 0.025 \times 2000 = 50$

$$P\{Y < 50\} = P\left\{\frac{Y - 2000 \times 0.02}{\sqrt{2000 \times 0.000075}} < \frac{50 - 2000 \times 0.02}{\sqrt{2000 \times 0.000075}}\right\} = \Phi(25.8)$$

? 试卷中没有给出 $\Phi(25.8)$ 的值, 且直观上感觉 $\Phi(25.8)$ 的值太大了, 故不能肯定题中的做

法是否可行

七、

$$(1) S_{\text{椭圆}} = \pi ab$$

$$\text{故 } (x, y) \text{ 的联合密度函数 } f(x, y) = \begin{cases} \frac{1}{\pi ab} & -a \leq x \leq a, -b \leq y \leq b \\ 0 & \text{其它} \end{cases}$$

$$X \text{ 的边缘密度函数 } f_X(x) = \begin{cases} \frac{2}{\pi a} & -a \leq x \leq a \\ 0 & \text{其它} \end{cases}$$

$$Y \text{ 的边缘密度函数 } f_Y(y) = \begin{cases} \frac{2}{\pi b} & -b \leq y \leq b \\ 0 & \text{其它} \end{cases}$$

$$(2) EX = \int_{-a}^a x \cdot \frac{2}{\pi a} dx = 0, EY = \int_{-b}^b y \cdot \frac{2}{\pi b} dy = 0,$$

$$EX^2 = \int_{-a}^a x^2 \cdot \frac{2}{\pi a} dx = \frac{4a^2}{3\pi}, EY^2 = \int_{-b}^b y^2 \cdot \frac{2}{\pi b} dy = \frac{4b^2}{3\pi}$$

$$DX = EX^2 - (EX)^2 = \frac{4a^2}{3\pi} = 25, DY = EY^2 - (EY)^2 = \frac{4b^2}{3\pi} = 4$$

$$\text{解得 } a = \frac{5\sqrt{3\pi}}{2}, b = \sqrt{3\pi}$$

(3) $-a \leq x \leq a, -b \leq y \leq b$ 时,

$$f_X(x) \cdot f_Y(y) = \frac{2}{\pi a} \cdot \frac{2}{\pi b} \neq \frac{1}{\pi ab}, \text{ 故 } X \text{ 与 } Y \text{ 不独立}$$

八、

$$\begin{aligned} Z \text{ 的分布函数 } F(z) &= P\{Z \leq z\} = 1 - P(Z > z) = 1 - P\{\min(X, Y) > z\} \\ &= 1 - P(X > z, Y > z) = 1 - P(X > z) \cdot P(Y > z) \end{aligned}$$

$$\text{当 } z \leq 0 \text{ 时, } P(X > z) = P(Y > z) = 1$$

$$\text{故 } F(z) = 1 - 1 = 0$$

$$\text{当 } 0 < z \leq 1 \text{ 时, } P(X > z) = \int_z^1 1 dx = 1 - z$$

$$\text{_____ } P(Y > z) = \int_z^1 5e^{-5x} dx = e^{-5z} - e^{-5}$$

$$\text{故 } F(z) = 1 - (1 - z) \cdot (e^{-5z} - e^{-5})$$

$$\text{当 } z > 1 \text{ 时, } P(X > z) = 0$$

$$\text{故 } F(z) = 1 - 0 = 1$$

$$\text{所以 } F(z) = \begin{cases} 0 & z \leq 0 \\ 1 - (1 - z) \cdot (e^{-5z} - e^{-5}) & 0 < z \leq 1 \\ 1 & z > 1 \end{cases}$$

$$f(z) = \begin{cases} 0 & z \leq 0 \\ 6e^{-5z} - 5ze^{-5z} - e^{-5} & 0 < z \leq 1 \\ 0 & z > 1 \end{cases}$$